THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 12	Number of pieces = $n \div$ piece length.
2	**(c)** 15	Scaling by a fraction is multiplication.
3	**(b)** 8	Use the meaning of multiplying and dividing fractions.
4	**(d)** 6	Convert to an improper fraction and multiply.
5	**(a)** $\frac{2}{3}$	Write as an improper fraction, then invert.
6	**(d)** 2	Multiply by the reciprocal of the divisor.
7	**(a)** $4\frac{1}{2}$	Multiply the quantity by the number of batches.
8	**(d)** 14	Number of pieces = $n \div$ piece length.
9	**(b)** 14	Scaling by a fraction is multiplication.
10	**(c)** 2	Use the meaning of multiplying and dividing fractions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Count = $\frac{3}{4} \div \frac{1}{8}$; = $\frac{3}{4} \times \frac{8}{1} = \frac{24}{4}$ (1); Answer 6 (1).	1+1
12	$\frac{3}{4}$ of 48= 36; $\frac{2}{3} \times 36$ (1); = 24 kg (1).	1+1
13	$x \times \frac{2}{5} = \frac{4}{15}$; $x = \frac{4}{15} \div \frac{2}{5}$ (1); $x = \frac{2}{3}$ (1).	1+1
14	Count = $\frac{5}{2} \div \frac{5}{6}$; = $\frac{5}{2} \times \frac{6}{5} = \frac{30}{10}$ (1); Answer 3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area = $\frac{3}{4} \times \frac{2}{5} = \frac{3}{10}$ m^2 (1); Smaller side squared = $\frac{4}{25}$ m^2 (1); Difference = $\frac{7}{50}$ m^2 (1).	1+1+1
16	Improper form $\frac{5}{2}$; $\frac{5}{2} \times \frac{4}{3}$ (1); = $\frac{20}{6}$ (1); Answer $3\frac{1}{3}$ (1).	1+1+1
17	Cloth for 8 shirts = $8 \times \frac{5}{4}$; = 10 m (1); Shirts from 10\frac{5}{8} m = $8\frac{1}{2}$ (1); Full shirts = 8 (1).	1+1+1
	OR: Area = $\frac{5}{6} \times \frac{3}{10} = \frac{1}{4}$ m^2 (1); Smaller side squared = $\frac{9}{100}$ m^2 (1); Difference = $\frac{4}{25}$ m^2 (1).	1+1+1
18	Improper form $\frac{5}{3}$; $\frac{5}{3} \times \frac{6}{5}$ (1); = $\frac{30}{15}$ (1); Answer 2 (1).	1+1+1
	OR: Cloth for 6 shirts = $6 \times \frac{5}{2}$; = 15 m (1); Shirts from 16\frac{1}{4} m = $6\frac{1}{2}$ (1); Full shirts = 6 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First part = $\frac{1}{4} \times 120 = 30$ km (1); Remaining = $120 - 30 = 90$ km (1); Second part = $\frac{2}{3} \times 90 = 60$ km (1); Left = $90 - 60$ (1); = 30 km (1).	1+1+1+1+1
	OR: Area = $\frac{4}{5} \times \frac{5}{6} = \frac{2}{3}$ (1); Vegetable part = $\frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ (1); Flower part = $\frac{2}{3} - \frac{1}{2}$ (1); = $\frac{1}{6}$ (1); State the three areas with units (1).	1+1+1+1+1
20	First part = $\frac{1}{3} \times 90 = 30$ km (1); Remaining = $90 - 30 = 60$ km (1); Second part = $\frac{3}{4} \times 60 = 45$ km (1); Left = $60 - 45$ (1); = 15 km (1).	1+1+1+1+1
	OR: Area = $\frac{3}{4} \times \frac{8}{9} = \frac{2}{3}$ (1); Vegetable part = $\frac{2}{3} \times \frac{2}{3} = \frac{4}{9}$ (1); Flower part = $\frac{2}{3} - \frac{4}{9}$ (1); = $\frac{2}{9}$ (1); State the three areas with units (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 14	Convert to an improper fraction and multiply.
2	**(b)** $\frac{4}{9}$	Write as an improper fraction, then invert.
3	**(d)** 2	Multiply by the reciprocal of the divisor.
4	**(b)** 6	Multiply the quantity by the number of batches.
5	**(d)** 12	Number of pieces $= n \div$ piece length.
6	**(c)** 15	Scaling by a fraction is multiplication.
7	**(c)** $\frac{1}{2}$	Use the meaning of multiplying and dividing fractions.
8	**(b)** 30	Convert to an improper fraction and multiply.
9	**(c)** $\frac{3}{11}$	Write as an improper fraction, then invert.
10	**(b)** 2	Multiply by the reciprocal of the divisor.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\frac{5}{6}$ of $36 = 30$; $\frac{3}{5} \times 30$ (1); $= 18$ litres (1).	1+1
12	$x \times \frac{3}{4} = \frac{9}{20}$; $x = \frac{9}{20} \div \frac{3}{4}$ (1); $x = \frac{3}{5}$ (1).	1+1
13	Count $= \frac{2}{3} \div \frac{1}{9}$; $= \frac{2}{3} \times \frac{9}{1} = \frac{18}{3}$ (1); Answer 6 (1).	1+1
14	$\frac{2}{7}$ of $70 = 20$; $\frac{1}{2} \times 20$ (1); $= 10$ metres (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{7}{8} \times \frac{4}{9} = \frac{7}{18}$ m^2 (1); Smaller side squared $= \frac{16}{81}$ m^2 (1); Difference $= \frac{31}{162}$ m^2 (1).	1+1+1
16	Improper form $\frac{10}{3}$; $\frac{10}{3} \times \frac{9}{5}$ (1); $= \frac{90}{15}$ (1); Answer 6 (1).	1+1+1
17	Cloth for 9 shirts $= 9 \times \frac{5}{3}$; $= 15$ m (1); Shirts from 15\frac{5}{6} m $= 9\frac{1}{2}$ (1); Full shirts $= 9$ (1).	1+1+1
	OR: Area $= \frac{2}{3} \times \frac{9}{10} = \frac{3}{5}$ m^2 (1); Smaller side squared $= \frac{4}{9}$ m^2 (1); Difference $= \frac{7}{45}$ m^2 (1).	1+1+1
18	Improper form $\frac{11}{4}$; $\frac{11}{4} \times \frac{15}{8}$ (1); $= \frac{165}{32}$ (1); Answer $5\frac{5}{32}$ (1).	1+1+1
	OR: Cloth for 12 shirts $= 12 \times \frac{9}{4}$; $= 27$ m (1); Shirts from 28\frac{1}{8} m $= 12\frac{1}{2}$ (1); Full shirts $= 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First part $= \frac{3}{8} \times 160 = 60$ km (1); Remaining $= 160 - 60 = 100$ km (1); Second part $= \frac{2}{5} \times 100 = 40$ km (1); Left $= 100 - 40$ (1); $= 60$ km (1).	1+1+1+1+1
	OR: Area $= \frac{5}{6} \times \frac{6}{7} = \frac{5}{7}$ (1); Vegetable part $= \frac{4}{5} \times \frac{5}{7} = \frac{4}{7}$ (1); Flower part $= \frac{5}{7} - \frac{4}{7}$ (1); $= \frac{1}{7}$ (1); State the three areas with units (1).	1+1+1+1+1
20	First part $= \frac{2}{7} \times 84 = 24$ km (1); Remaining $= 84 - 24 = 60$ km (1); Second part $= \frac{3}{5} \times 60 = 36$ km (1); Left $= 60 - 36$ (1); $= 24$ km (1).	1+1+1+1+1
	OR: Area $= \frac{7}{8} \times \frac{4}{5} = \frac{7}{10}$ (1); Vegetable part $= \frac{5}{7} \times \frac{7}{10} = \frac{1}{2}$ (1); Flower part $= \frac{7}{10} - \frac{1}{2}$ (1); $= \frac{1}{5}$ (1); State the three areas with units (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $2\frac{1}{2}$	Multiply the quantity by the number of batches.
2	**(a)** 8	Number of pieces $= n \div$ piece length.
3	**(d)** 21	Scaling by a fraction is multiplication.
4	**(a)** $4 \div \frac{1}{2}$	Use the meaning of multiplying and dividing fractions.
5	**(c)** 7	Convert to an improper fraction and multiply.
6	**(b)** $\frac{5}{8}$	Write as an improper fraction, then invert.
7	**(b)** $1\frac{1}{2}$	Multiply by the reciprocal of the divisor.
8	**(d)** $2\frac{1}{2}$	Multiply the quantity by the number of batches.
9	**(b)** 10	Number of pieces $= n \div$ piece length.
10	**(b)** greater than the number	Use the meaning of multiplying and dividing fractions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x \times \frac{5}{6} = \frac{25}{36}$; $x = \frac{25}{36} \div \frac{5}{6}$ (1); $x = \frac{5}{6}$ (1).	1+1
12	Count $= \frac{7}{4} \div \frac{7}{16}$; $= \frac{7}{4} \times \frac{16}{7} = \frac{112}{28}$ (1); Answer 4 (1).	1+1
13	$\frac{8}{9}$ of $54 = 48$; $\frac{3}{4} \times 48$ (1); $= 36$ rupees (1).	1+1
14	$x \times \frac{7}{8} = \frac{21}{40}$; $x = \frac{21}{40} \div \frac{7}{8}$ (1); $x = \frac{3}{5}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Area $= \frac{5}{8} \times \frac{4}{15} = \frac{1}{6}$ m^2 (1); Smaller side squared $= \frac{16}{225}$ m^2 (1); Difference $= \frac{43}{450}$ m^2 (1).	1+1+1
16	Improper form $\frac{5}{4}$; $\frac{5}{4} \times \frac{12}{5}$ (1); $= \frac{60}{20}$ (1); Answer 3 (1).	1+1+1
17	Cloth for 10 shirts $= 10 \times \frac{7}{4}$; $= 17\frac{1}{2}$ m (1); Shirts from 18\frac{3}{8} m $= 10\frac{1}{2}$ (1); Full shirts $= 10$ (1).	1+1+1
	OR: Area $= \frac{3}{7} \times \frac{14}{15} = \frac{2}{5}$ m^2 (1); Smaller side squared $= \frac{9}{49}$ m^2 (1); Difference $= \frac{53}{245}$ m^2 (1).	1+1+1
18	Improper form $\frac{9}{2}$; $\frac{9}{2} \times \frac{8}{3}$ (1); $= \frac{72}{6}$ (1); Answer 12 (1).	1+1+1
	OR: Cloth for 9 shirts $= 9 \times \frac{10}{3}$; $= 30$ m (1); Shirts from 31\frac{2}{3} m $= 9\frac{1}{2}$ (1); Full shirts $= 9$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First part $= \frac{2}{5} \times 150 = 60$ km (1); Remaining $= 150 - 60 = 90$ km (1); Second part $= \frac{5}{6} \times 90 = 75$ km (1); Left $= 90 - 75$ (1); $= 15$ km (1).	1+1+1+1+1
	OR: Area $= \frac{9}{10} \times \frac{5}{6} = \frac{3}{4}$ (1); Vegetable part $= \frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$ (1); Flower part $= \frac{3}{4} - \frac{3}{10}$ (1); $= \frac{9}{20}$ (1); State the three areas with units (1).	1+1+1+1+1
20	First part $= \frac{3}{10} \times 200 = 60$ km (1); Remaining $= 200 - 60 = 140$ km (1); Second part $= \frac{4}{7} \times 140 = 80$ km (1); Left $= 140 - 80$ (1); $= 60$ km (1).	1+1+1+1+1
	OR: Area $= \frac{2}{3} \times \frac{3}{8} = \frac{1}{4}$ (1); Vegetable part $= \frac{5}{6} \times \frac{1}{4} = \frac{5}{24}$ (1); Flower part $= \frac{1}{4} - \frac{5}{24}$ (1); $= \frac{1}{24}$ (1); State the three areas with units (1).	1+1+1+1+1

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